SEMICONDUCTOR DESIGN

Multi-Process Synthesis & Timing Closure

CISC Micro-Processor · Samsung 32nm · TSMC 180nm · 130nm

RTL-to-Netlist synthesis, register-retiming, formal equivalence (Formality), and cross-process timing closure.

정태웅 · 청주대학교

Design Compiler · Formality · VCS

CONTENTS

Agenda

Problem → analysis → fix → verification, applied across three process nodes.

01	RTL & Foundations	CISC datapath, schematics, functional simulation, multi-Vt libraries, and the design spec.
02	Samsung 32nm	Hierarchy, the Formality temp_reg fix, the half-cycle path, and timing closure (main focus).
03	TSMC 180nm · 130nm	Cross-library migration (capacitance units), topographical vs WLM, and process-node limits.
04	Verification & Closing	Cross-process comparison, Formality equivalence, netlist simulation, and key takeaways.

Reference: Design2 (CISC) Specification · 정태웅 · 청주대학교

OVERVIEW

Project Overview

One RTL design synthesized and timing-closed across three nodes, verified at every stage.

3	1.4 ns	3	PASS
Process Nodes	**Target Period**	**Verification Layers**	**Formality**
Samsung 32nm · TSMC 180nm · TSMC 130nm from one RTL	714 MHz spec; met at 32nm, floor 1.29 ns explored	RTL · post-synthesis netlist · gate-level VCS	RTL ≡ netlist confirmed after temp_reg fix

Scope at a glance

Baseline (32nm)

Samsung SAED32 (topographical) meets 714 MHz with 0 timing / 0 DRC violations — the compliant reference.

Methodology

Hierarchy-preserving compile_ultra, register retiming, and a Formality-driven dont_touch fix for a removed register.

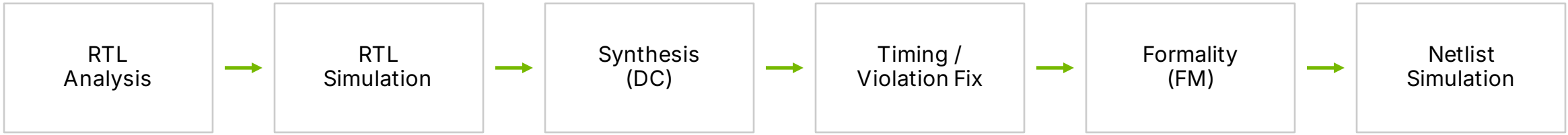
Cross-process

The same RTL ported to slower TSMC nodes exposes process-speed limits after correct capacitance-unit migration.

OVERVIEW

Common Design Methodology

A unified RTL-to-Netlist flow applied identically to every process node.



RTL Code Analysis

Interpret the CISC hierarchy (MC · memory · bidir_core2 · clk_gen), sketch the datapath, and locate critical paths.

Hierarchical Synthesis

compile_ultra -no_autoungroup keeps module boundaries; -retime relocates registers to improve delay.

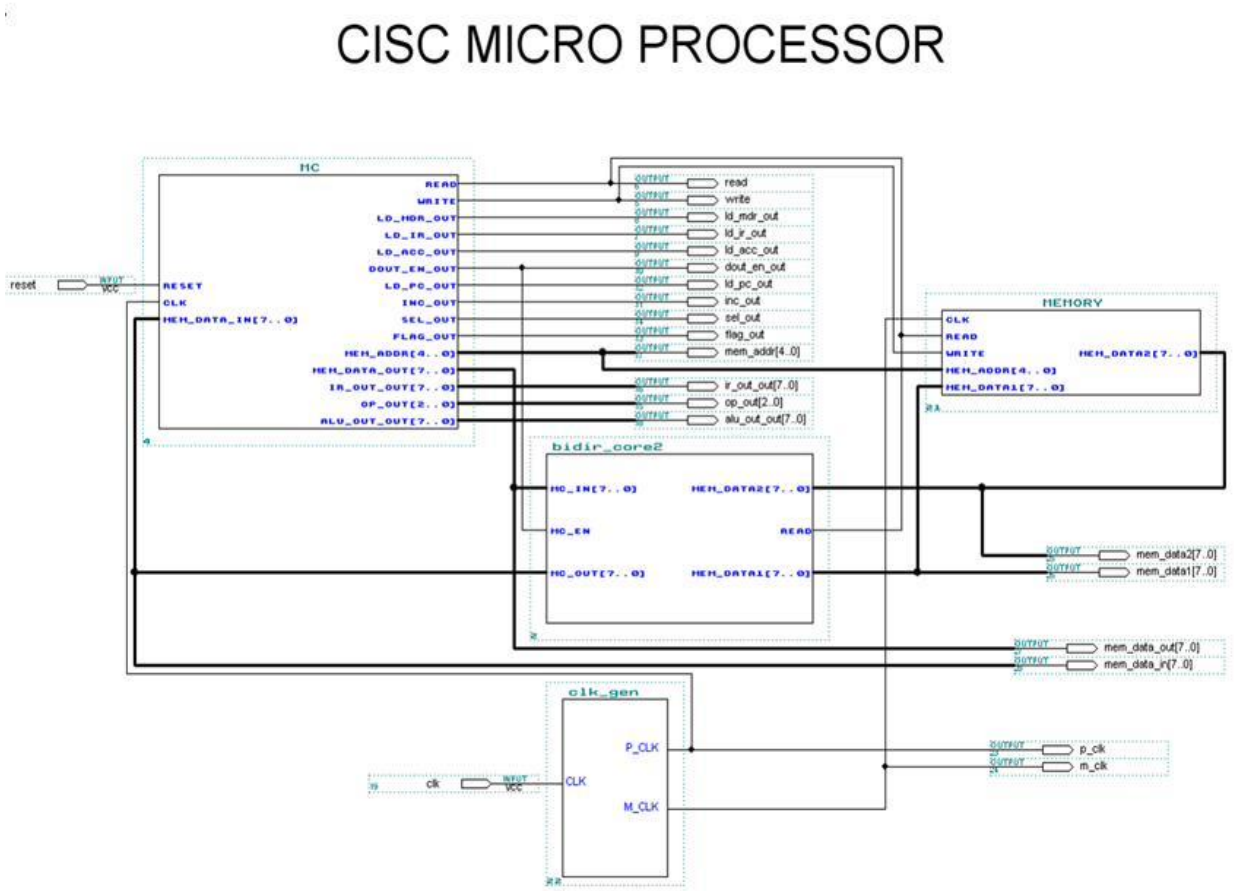
Equivalence & Sim

Formality confirms RTL ≡ netlist; gate-level VCS reproduces the RTL response at each period.

Reference: Synopsys Synthesis Tool Commands · 정태웅 · 청주대학교

CISC Architecture— Top Level

An 8-bit CISC micro-processor: controller, memory, bidirectional bus core, and a dual-phase clock.



MC

Controller / datapath: MDR, IR, ALU, ACC, FLAG, STATE, PC_LOGIC, ADDR_MUX.

MEMORY

5-bit addressed store; read / write driven by MC; outputs mem_data1/2.

bidir_core2

Bidirectional bus interface between memory data and the controller.

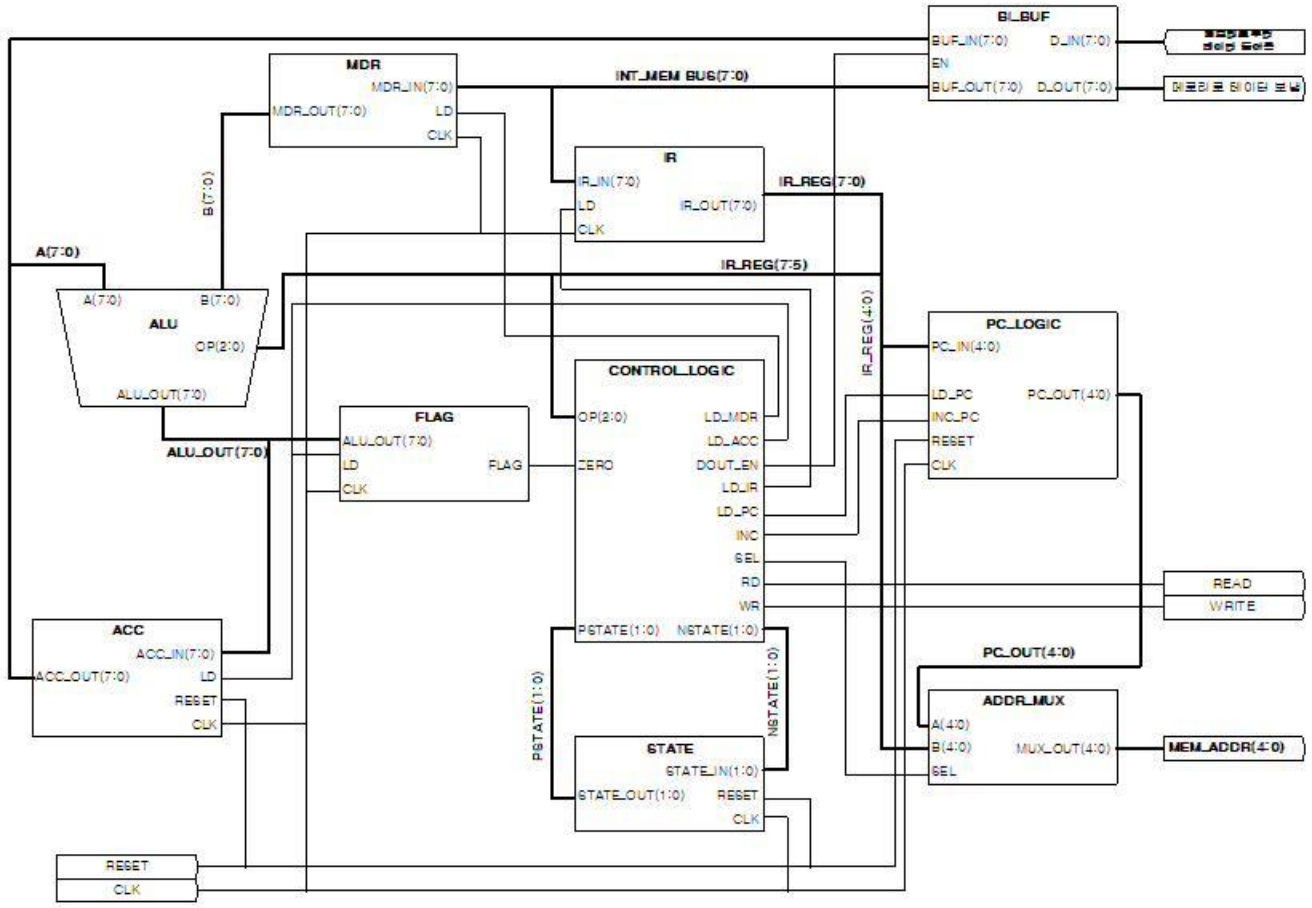
clk_gen

Generates p_clk / n_clk phases (the clk and clk' path groups).

Reference: CISC RTL (cisc.v) · 정태웅 · 청주대학교

Controller Datapath (MC)

Inside MC — the state machine, program counter, ALU/accumulator, and register file feeding the bus.

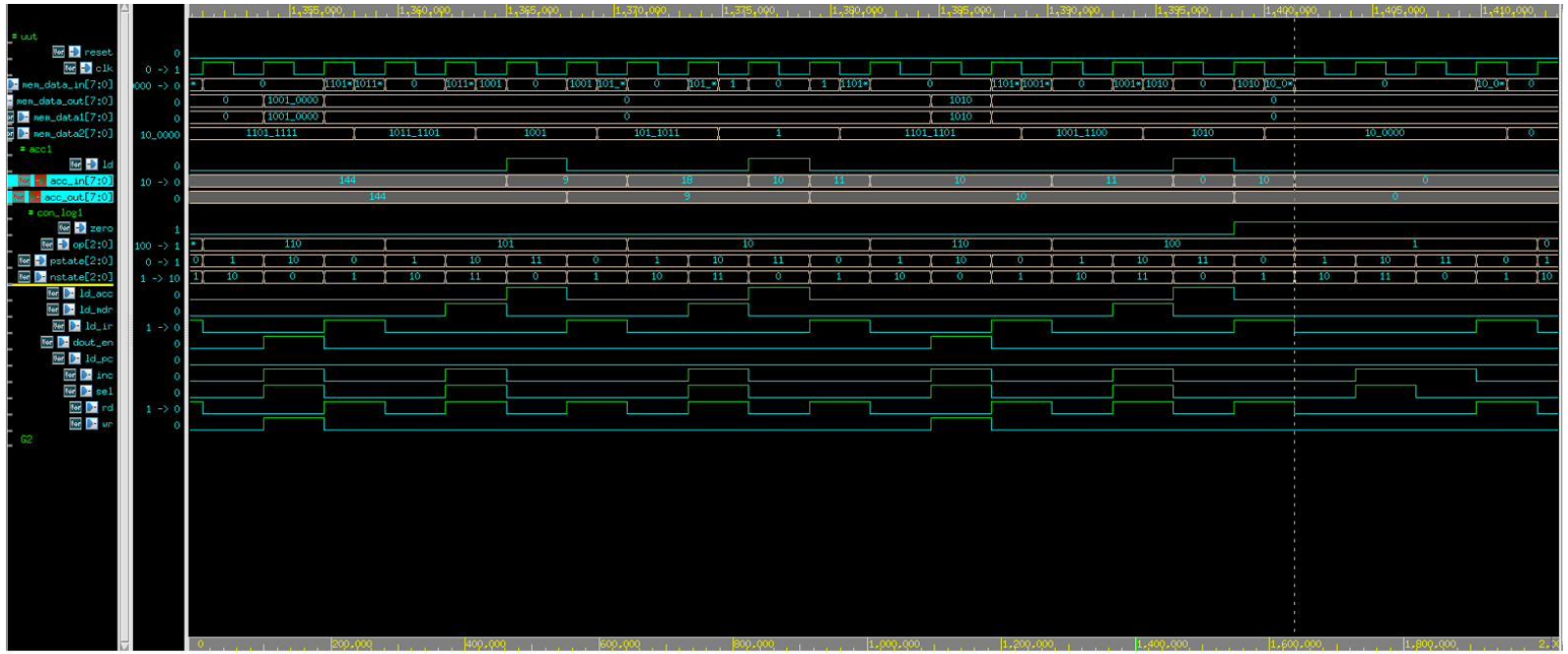


Key blocks

- **CONTROL_LOGIC / STATE**
  Decodes IR, sequences PSTATE → NSTATE, drives all load / enable signals.
- **ALU · ACC · FLAG**
  8-bit arithmetic with accumulator and zero flag.
- **PC_LOGIC · ADDR_MUX**
  Program counter update and memory-address selection.
- **MDR · IR · BI_BUF**
  Data / instruction registers on the shared INT_MEM_BUS.

RTL Functional Verification

Behavioral VCS simulation confirms controller and memory behavior against the testbench vectors.



Reset → fetch → decode → execute is exercised over the memory address range; mem_data1/2 and the bus signals match the expected sequence for each instruction. This RTL response is the golden reference for the later gate-level netlist simulations.

Design Spec & Constraints

Every constraint traces to the Design2 (CISC) specification — not arbitrary numbers.

Clock

- 714 MHz → 1.4 ns period
- Insertion 150 ± 5 ps · ext-gen delay 100 ps
- Jitter ± 10 ps · setup margin 20 ps
- Uncertainty ≥ 0.02 ns · transition 140 ps

I/O timing

- reset* input path ≤ 960 ps · min ext 0.02 ns
- mem_* output path ≤ 860 ps · min ext −0.02 ns
- Register setup ≤ 0.1 ns

Design rules

- max_fanout 40 · max_capacitance 5
- mem_* load = 6 × MUX41X1_HVT / A1 pin
- A1 cap 0.434962 fF → load ≈ 2.61 fF

Area & Vth

- LVT usage < 25 % of the design
- Boundary shape L · {{0 0}...{72 0}}
- Multi-Vt: HVT / RVT / LVT libraries

Spec Recommendation

The recommendation sheet defines the acceptance targets for this synthesis stage.

Criterion	Target	Notes
Setup — Reg-to-Reg	0 violations	Internal register paths must fully meet.
Setup — Input paths	0 violations	Paths to input ports must meet.
Setup — Output paths	WNS ≥ −0.05 ns	Small residual on output paths tolerated.
Hold	Monitor only	Resolved later during P&R.
DRC (trans / cap / fanout)	0 violations	Strict compliance with library limits.
Power — Vth	LVT ≤ 25 % (28 % ok)	Up to 28 % LVT allowed for timing.

compile_ultra — Key Options & Settings

Which option/command drove each output — the same toolkit, tuned per run.

Option / setting	flat	hier 1.4ns	1.29ns	T-180	T-130	Purpose
`compile_ultra_ungroup_dw = false`	X	O	O	O	O	Keep DesignWare blocks grouped.
`compile_timing_high_effort = true`	X	O	O	O	O	Extra timing-focused optimization.
`set_optimize_registers true`	X	O	X	O	O	Global retiming (can delete a register).
`-no_autoungroup`	X	O	O	O	O	Preserve the logical hierarchy.
`-only_map → set_dont_touch`	X	O	X	O	O	Map first, then protect temp_reg.
`-retime`	X	O	O	O	O	Relocate registers to cut delay.
`create_auto_path_groups`	X	O	O	O	O	Build group_path per endpoint to focus timing optimization.

O used in that run X not used. Global retiming (optimize_registers) is exactly where only_map + dont_touch is required.

Reference: Synopsys Synthesis Tool Commands · 정태웅 · 청주대학교

Formality Fails — a Register Vanished

With global retiming enabled, mem1/temp_reg is removed and its logic no longer matches the RTL.

```
**************************************************************************************
********


******************************** Verification Results ************************
********
Verification FAILED
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
---------------------------------------------------
 Reference design: r:/WORK/cisc
 Implementation design: i:/WORK/cisc
 151 Passing compare points
 20 Failing compare points (20 matched, 0 unmatched)
 0 Aborted compare points
 41 Unverified compare points
--------------------------------------------------------------------------------
--------
Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT
   TOTAL
--------------------------------------------------------------------------------
--------
Passing (equivalent)          85       0       0       0      32      34       0
     151
Failing (not equivalent)       0       0       0       0       0      20       0
      20
Unverified                     0       0       0       0       0      41       0
      41
**************************************************************************************
```

fm_shell Verification Results — 20 failing (DFF) + 41 unverified compare points.

VERIFICATION FAILED

- **What happened**

  set_optimize_registers true turns -retime into global retiming; DC removes mem1/temp_reg as redundant.

- **Why dont_retime wasn't enough**

  set_dont_retime blocks moving a register, not deleting it — so temp_reg still disappeared.

- **Result**

  20 failing DFF compare points and 41 unverified → RTL ≠ netlist. Equivalence cannot pass.

Fix — only_map, then dont_touch

Map first so the register exists, protect it with dont_touch, then retime incrementally.

compile_ultra -only_map → set_dont_touch mem1/temp_reg → compile_ultra -retime -incremental → Verification Succeeded

VERIFICATION SUCCEEDED

```
****************************** Verification Results ******************************
Verification SUCCEEDED
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
----------------------------------------------------------
 Reference design: r:/WORK/cisc
 Implementation design: i:/WORK/cisc
 122 Passing compare points
--------------------------------------------------------------------------------
Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT   TOTAL
--------------------------------------------------------------------------------
Passing (equivalent)           0       0       0       0      32      90       0     122
Failing (not equivalent)       0       0       0       0       0       0       0       0
********************************************************************************
```

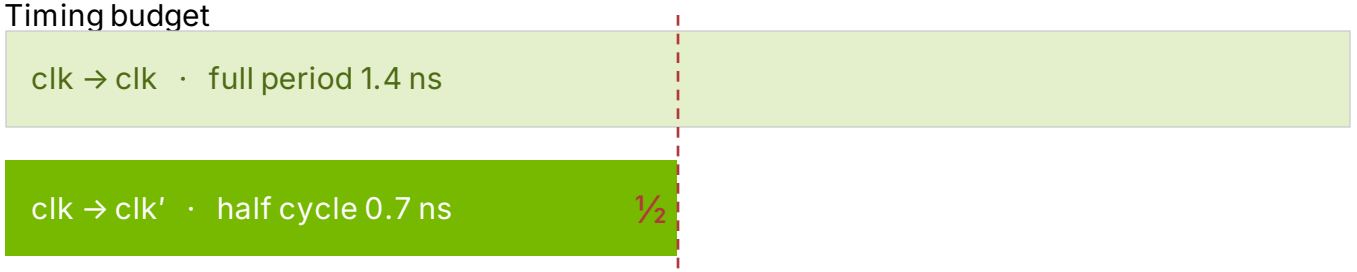
fm_shell Verification Results — 122 passing, 0 failing compare points.

Why only_map is needed: setting dont_touch before mapping blocks the map itself. -only_map runs mapping alone so temp_reg exists as a real cell, then dont_touch preserves it through all later optimization. (-only_map is a DC NXT option, absent from standard DC's man page, yet it ran correctly here.)

The Half-Cycle Path — clk′ Domain

The memory uses a negative-edge clock, so the tightest path has only half the period to close.

Timing budget



Negative-edge memory

The memory registers latch on the inverted clock (clk′), 0.7 ns out of phase.

Only half the period

A clk → clk′ path must arrive within 0.7 ns, not 1.4 ns — the true timing bottleneck.

It is the critical path

This clk′ endpoint (mem1/xn1_reg) sets the whole 32nm slack: +0.000147 ns at 1.4 ns.

Critical path (report_timing)

Startpoint	top_mc1/state1/state_out_reg[2]	
Endpoint	mem1/xn1_reg[6]	
Path Group	clk	
data arrival	0.857771	
clock clk′ (rise)	0.700000	← half
setup time	−0.052082	
data required	0.857918	
slack (MET)	0.000147	

At 1.29 ns the same clk′ path scales to 0.645 ns — the half-cycle limit follows the period down.

Period Floor — 1.29 ns vs 1.28 ns

Because of the half-cycle path, 1.29 ns is the tightest period that still holds; 1.28 ns breaks it.

Period 1.28 ns

MISSES TARGET

Reg-to-reg	MET
Output WNS	−0.051 ns
Output violations	16 paths
vs target (−0.05)	EXCEEDS → FAIL

Period 1.29 ns

MEETS TARGET

Reg-to-reg	MET (+0.001)
Output WNS	−0.048 ns
Output violations	16 paths
vs target (−0.05)	WITHIN → PASS

Trade-off: tightening 1.4 ns → 1.29 ns pushes LVT usage 24.8 % → 26.9 % — over the 25 % target but within the 28 % allowance the recommendation grants for timing. Faster costs leakage.

Samsung 32nm — Results by Run

Flat was an intermediate step; the hierarchical 1.4 ns and 1.29 ns netlists are the deliverables.

	Flat (intermediate)	Hier · 1.4 ns	Hier · 1.29 ns
Timing	1.4 ns MET	1.4 ns MET	1.29 ns · out −0.048
Cell area (µm²)	2 360	2 231	2 355
Leaf / Seq cells	697 / 168	846 / 90	776 / 135
Buf / Inv cells	132	227	181
LVT usage	23.7 %	24.8 %	26.9 %
Total power (mW)	1.37	1.12	1.38
Internal observability	no	yes	yes
Formality	PASS	PASS	PASS

Flat is dropped from delivery (internal signals unobservable). The 1.29 ns run trades leakage for speed at the half-cycle floor.

Library Migration — Capacitance Units

The output load was a Samsung cell reference; for TSMC it is read from the library and set in pF.

Item	SAED32	TSMC tcb018
Library header	capacitive_load_unit = ff	capacitance = pf
Capacitance unit	fF	**pF (1 pF = 1000 fF)**
mem_* output load	6 × MUX41X1_HVT / A1 pin (0.434962 fF) ≈ 2.61 fF	**pin cap from TSMC .lib → explicit numeric load (pF)**
Timing / transition	unit-independent	left untouched

Migration — done up front

1

Load was defined as 6 × a Samsung cell pin — invalid across libraries.

2

Read the cell's pin capacitance from the TSMC .lib and set an explicit numeric load.

3

Match the unit: SAED32 uses fF, TSMC uses pF (1 pF = 1000 fF).

4

Timing / transition are unit-independent — left untouched.

TSMC Results at 1.4 ns

Units correctly migrated and Formality passing — yet both nodes miss 714 MHz on speed alone.

TSMC 180nm

FM PASS · VIOLATED

Topographical · tcb018g3d3wc

WNS	−1.936 ns
TNS	−153.8 ns
Violating paths	120
Cell area (µm²)	30 685
Leaf cells	1138
Buf / Inv cells	343

TSMC 130nm

FM PASS · VIOLATED

WLM (wire-load) · scx2_tsmc_cl013g_ss

WNS	−1.213 ns
TNS	−135.4 ns
Violating paths	166
Cell area (µm²)	8 737
Leaf cells	667
Buf / Inv cells	150

Same RTL and flow (only_map + dont_touch + retime). 130nm had no tech DB, so a wire-load model was used — its RC is fanout-estimated, not placement-based.

Reference: Design Compiler QoR · 정태웅 · 청주대학교

Why the Nodes Differ

Same design; the 32nm ↔ 180/130nm gap is process scaling — and 130nm's numbers are estimates.

Feature size

Larger geometry → bigger cells. Area grows 2 231 µm² (32nm) → 8 737 (130nm) → 30 685 (180nm).

Transistor speed

Smaller nodes switch faster. 32nm clears 1.4 ns; 130/180nm cannot, leaving WNS −1.2 / −1.9 ns.

Caveat — 130nm ran a Wire-Load Model (WLM)

With no tech DB, 130nm could not use topographical (placement-based) RC. WLM estimates interconnect delay from fanout and wire-load tables only — no real placement. So 130nm's area, RC, and timing are approximations, not directly comparable to the placement-based 32nm and 180nm topographical results. 180nm and 32nm, having tech DBs, are the like-for-like pair.

Cross-Process Comparison

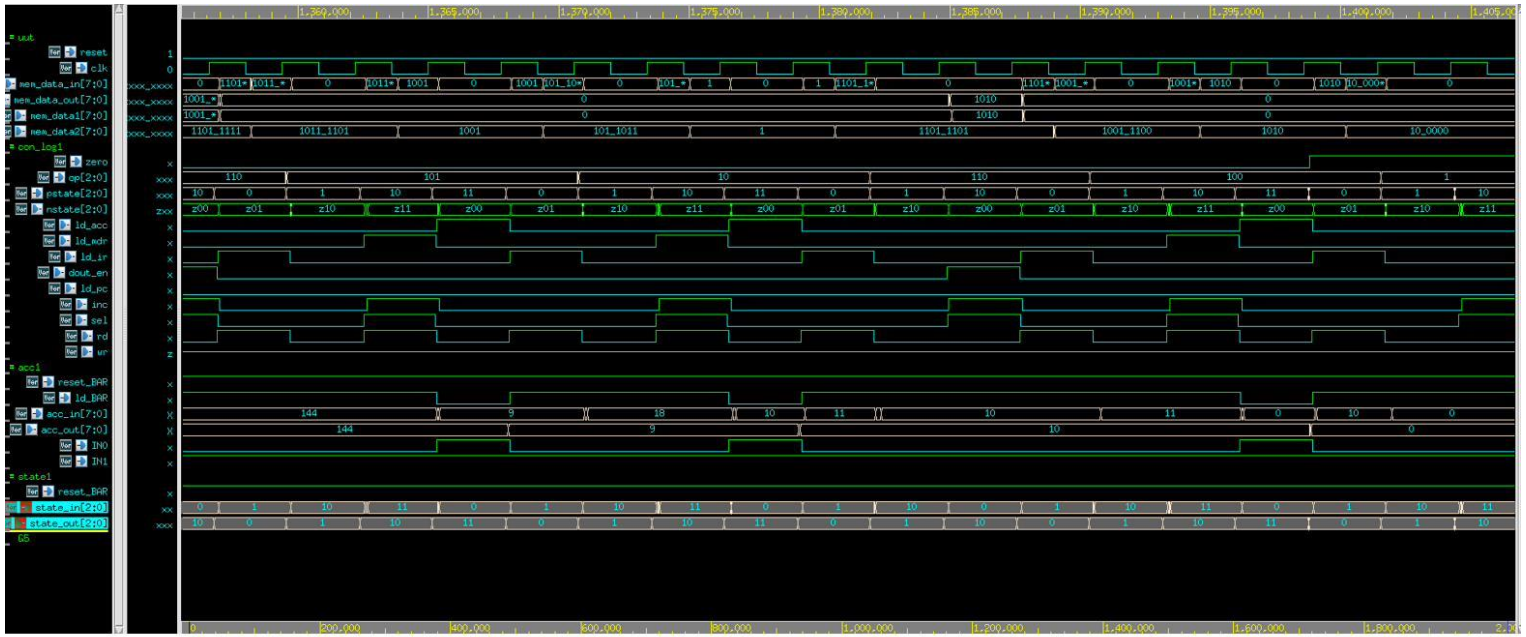
Baseline = Samsung 32nm hierarchical at 1.4 ns; the table shows what actually differs.

	Samsung 32nm	TSMC 130nm	TSMC 180nm
Mode	Topographical	WLM (est.)	Topographical
Timing @ 1.4 ns	MET · WNS 0	−1.213 ns	−1.936 ns
Violating paths	0	166	120
Cell area (µm²)	2 231	8 737 (est.)	30 685
Leaf cells	846	667	1138
Buf / Inv cells	227	150	343
Formality	PASS	PASS	PASS

Shared: identical RTL · -no_autoungroup · only_map + dont_touch(temp_reg) · retime. 130nm uses a wire-load model, so "(est.)" values are fanout-based approximations — not placement-based.

Netlist Simulation — 1.4 ns

The gate-level netlist reproduces the RTL functional result at the 1.4 ns target period.



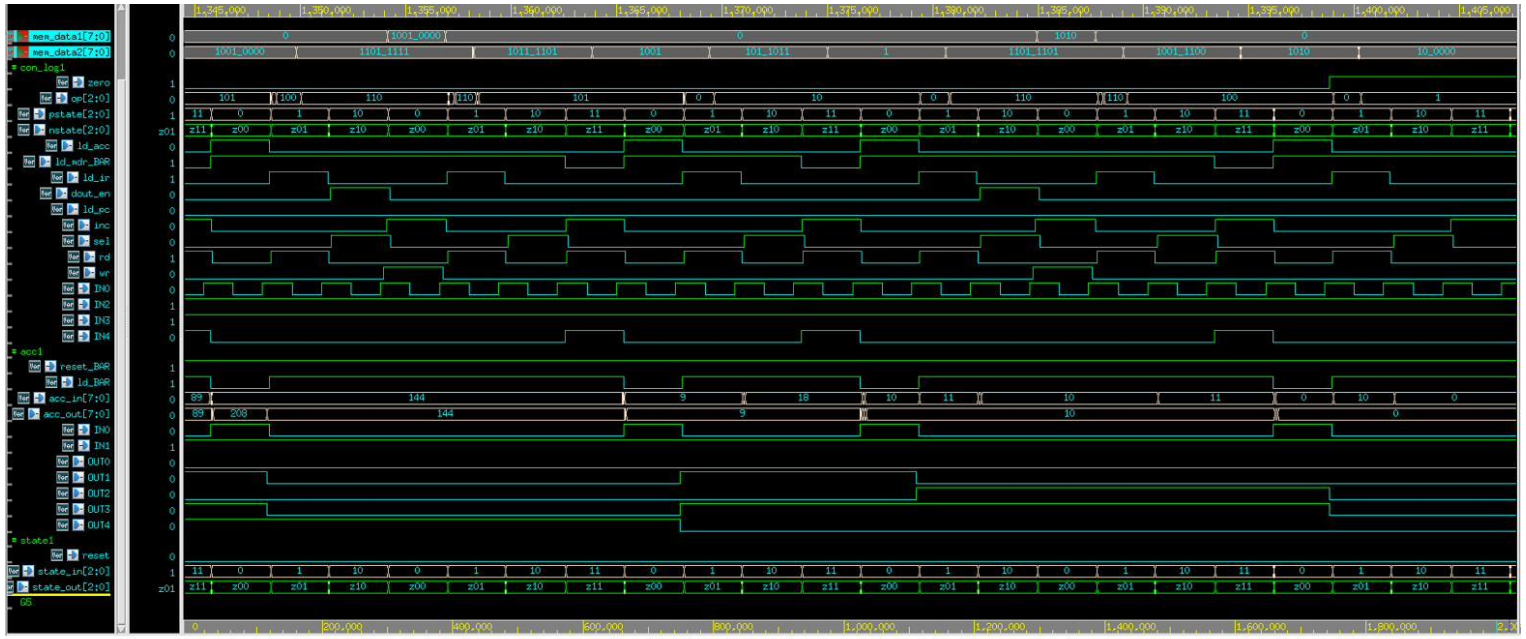
Samsung 32nm gate-level netlist · normal operation

What the waveform shows

- Final result settles to 144 — the expected program output.
- The zero flag goes high, signalling that the program has finished.
- After synthesis, the nstate reset-decision bit reads z (high-Z) — this bit is pulled out as reset_bar (active-high reset).

Netlist Simulation — 1.29 ns

At the tightest passing period, the netlist still matches the RTL behavior.



Samsung 32nm gate-level netlist · normal operation

What the waveform shows

- Functionally identical to the 1.4 ns run — same 144 result and zero-flag termination.
- Confirms that closing at the half-cycle floor (1.29 ns) preserves correct behavior.
- After synthesis, the nstate reset-decision bit reads z (high-Z) — this bit is pulled out as reset_bar (active-high reset).

Key Takeaways

01

Retiming and equivalence are co-designed

Register retiming can silently delete a register (mem1/temp_reg) and break Formality. It must be paired with a preservation strategy — only_map → dont_touch → recompile — or equivalence fails.

02

Hierarchy eases functional verification

Preserving module boundaries keeps internal signals visible in VCS, so predicting and checking behavior is far easier than on a flattened netlist that exposes only I/O.

03

Timing closure has a quantified cost

The half-cycle clk' path makes 1.29 ns the 32nm floor, and that speed costs leakage: LVT rises 24.8 % → 26.9 % (within the 28 % allowance).

04

The node sets the ceiling

32nm meets 714 MHz; 130/180nm cannot — an honest process limit. 130nm's WLM numbers are estimates, not a like-for-like comparison.

VERIFICATION & CLOSING

One RTL, three nodes, verified end to end.

Next steps — trim LVT on the half-cycle path to reclaim leakage margin, and carry the 32nm netlist into place-and-route.

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Thank you